SafeChat: A Grooming-Aware Chat Application for Female Online Safety

"An AI That Reads Between The Lines"

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Objectives

The objectives of this project are as follows:

- To design and develop SafeChat, a purpose-built AI-incorporated native Android chat application that embeds real-time grooming risk detection within a native messaging environment for female users of all ages.
- To implement a weighted keyword scoring system that analyses outgoing messages in real time, classifying grooming risk into three actionable levels: Safe, Warning, and Blocked.
- To integrate the Google Gemini API within a dedicated Safe Hub feature, providing users with an AI-powered conversational assistant for grooming awareness, safety education, and guidance.
- To provide users with a per-contact risk dashboard that aggregates grooming-related behaviour over time, allowing users to assess the overall risk profile of individuals they are communicating with.
- To raise awareness among female users regarding the subtle early stages of grooming behaviour through intelligent in-app feedback and an accessible AI safety assistant.
- To design the system with scalability in mind, allowing future adoption by child welfare organisations, NGOs, and digital safety initiatives globally.

Problem Statement

Online grooming is a form of sexual exploitation in which perpetrators systematically build trust with potential victims, typically through digital communication platforms, with the intent of facilitating sexual abuse or harassment. Unlike overt forms of harassment, grooming operates through gradual, seemingly benign interactions: excessive compliments, emotional dependency, requests for secrecy, and progressive boundary violations. This subtlety makes it particularly dangerous, as victims often fail to recognise warning signs until significant harm has already occurred.

Females across all age groups represent a disproportionately high risk demographic for online grooming. Research consistently identifies women and girls as the primary targets of grooming behaviour, with perpetrators exploiting emotional vulnerability, social inexperience, and trust. The widespread adoption of social media platforms, online gaming communities, messaging applications, and video sharing services has created numerous channels through which groomers operate, reaching potential victims at increasingly younger ages and across geographical boundaries.

Current reporting mechanisms for online grooming and digital sexual harassment are often inaccessible, intimidating, or unknown to those who need them most. Victims frequently hesitate to report incidents, particularly when they are uncertain whether the behaviour they experienced constitutes a genuine threat. This gap between experience and awareness creates a critical need for a low-barrier tool that empowers users to assess their own situations independently, without requiring immediate formal disclosure or external intervention.

Critically, grooming does not occur in isolation. It unfolds within chat environments, through messaging platforms that form the core of daily digital social interaction for females of all ages. Existing AI-driven detection systems predominantly focus on platform-level content moderation, requiring specialised infrastructure and third-party access that places the burden of protection on institutions rather than individuals. There is a notable absence of tools that meet users where the risk actually occurs: within the conversation itself.

SafeChat addresses this gap by embedding AI-assisted grooming risk detection directly within a purpose-built native Android chat application. The system operates at two levels: a real-time keyword scoring engine that intercepts grooming-related messages before they are sent, and a Gemini-powered Safe Hub that provides users with on-demand safety education and guidance without leaving the app. The problem is therefore twofold: (1) females globally lack the awareness and tools to independently recognise grooming behaviour in online conversations, and (2) existing digital safety initiatives fail to integrate AI-driven risk assessment within the environments where grooming actually takes place. SafeChat is proposed as a direct response to both dimensions of this problem.

Contribution of the Project

This project contributes to the domain of digital safety and AI assisted welfare systems in the following ways:

Real-Time Message Interception: SafeChat's weighted keyword scoring system analyses every outgoing message before it is delivered, providing immediate in-app feedback to senders through warning alerts or message blocking. This proactive approach addresses grooming at the point of occurrence rather than after the fact.

Awareness Tool: The Safe Hub feature serves as an educational and awareness platform, powered by the Google Gemini API. Female users of all ages can interact with an AI assistant to understand grooming behaviour, seek safety guidance, and access actionable advice, empowering users with knowledge before harm occurs.

Accessible Risk Assessment: The per-contact risk dashboard provides users with an aggregated risk score for each individual they are communicating with, based on the cumulative grooming-related language detected in that conversation. This gives users a longitudinal view of risk without requiring formal disclosure or external intervention.

AI Application in Social Good: The project demonstrates a practical application of rule-based NLP and large language model integration in addressing real-world social harm, contributing to the growing body of work on AI for digital safety and humanitarian purposes.

Ethical AI Design: The in-chat report feature allows users to flag a conversation directly from the chat screen. Submitted conversation evidence is forwarded to the Gemini API for AI-assisted analysis, providing users with an informed summary before formal reporting. The final submission pipeline is designed for future integration with relevant authorities or support organisations.

Research Question

This project seeks to answer the following question:

“How effectively can a weighted keyword scoring system identify linguistic indicators of online grooming in real-time text-based conversations, and how can it be combined with a large language model assistant to deliver an integrated, on-demand safety tool within a native mobile chat application? ”

Keyword Lexicon Sources

The weighted keyword lexicon used by the real-time detection engine was informed by the following sources:

- Established grooming behaviour taxonomies from criminology and psychology literature, covering indicators such as solicitation of explicit content, requests for secrecy, emotional manipulation, and boundary escalation language.
- The PAN 2012 Sexual Predator Identification Dataset, used as a reference to validate the relevance and coverage of selected keywords against real labelled grooming conversations.
- Synthetic test conversations developed by the research team to verify system behaviour across Safe, Warning, and Blocked threshold scenarios.

Methodology

System Pipeline

Real-Time Detection Pipeline: Outgoing Message → Keyword Scoring Engine (Token Accumulation) → Threshold Classification → Safe (below 20 tokens) / Warning (20–49 tokens) / Blocked (50+ tokens) → In-App Alert or Block UI

Safe Hub Pipeline: User Query → Google Gemini API (Grooming-Informed System Prompt) → AI Safety Guidance Response → Safe Hub Chat UI

Report Pipeline: User triggers Report from chat screen → Conversation evidence forwarded to Gemini API → AI-assisted analysis returned → Report summary presented to user

AI Techniques

Weighted Keyword Scoring A curated lexicon of grooming-associated terms and phrases, derived from established criminological research, is applied to each outgoing message at the point of composition. Each keyword or phrase carries a predefined token weight reflecting its severity as a grooming indicator. Token values are accumulated per message and classified against two thresholds: a warning threshold at 20 tokens and a blocking threshold at 50 tokens. This approach ensures full transparency and explainability, as every risk classification is directly traceable to specific detected terms.

Threshold-Based Risk Classification The accumulated token score determines the system response:

- Below 20 tokens — message is sent normally (Safe)
- 20 to 49 tokens — sender receives a warning popup; message can still be sent (Warning)
- 50 tokens and above — message is blocked and cannot be sent (Blocked). The sender may submit an appeal for human review to account for legitimate contextual exceptions.

Cumulative Risk Dashboard Token scores are aggregated over the history of a conversation to produce a per-contact risk score, visible to the user through the Risk Dashboard. This longitudinal view allows users to assess whether a contact's overall communication pattern reflects escalating grooming behaviour over time, beyond the assessment of any single message.

Large Language Model Integration (Safe Hub) The Safe Hub is a dedicated screen within the application that provides an AI-powered conversational assistant using the Google Gemini API. The assistant is configured with a grooming-informed system prompt and responds to user queries about grooming behaviour, warning signs, and safety guidance. This component operates independently of the real-time detection pipeline and serves primarily as an awareness and education tool.

Critical Review of Algorithm Suitability

The weighted keyword scoring approach is selected for its transparency and explainability, which are critical in a sensitive domain such as harassment detection. Unlike black-box deep learning models, the token-based system allows developers and users to trace exactly which terms triggered a given classification, supporting user trust and ethical accountability.

However, limitations exist. Keyword-based systems may be circumvented by novel grooming tactics that fall outside the defined lexicon, including deliberate misspellings, coded language, or multilingual expression. The system does not perform contextual disambiguation at the detection layer, meaning that legitimate uses of flagged terms may trigger warnings or blocks.

The appeal mechanism is introduced to partially address this limitation by enabling human review of contested cases.

The Gemini-powered Safe Hub component introduces a complementary layer of contextual language understanding, though it operates on user-initiated queries rather than passive message monitoring. Future iterations may incorporate supervised machine learning classifiers or contextual NLP models to improve detection accuracy and reduce false positives at the message interception layer.

Technology Stack

- Frontend: Native Android application developed in Java using Android SDK, with View Binding for UI interaction and RecyclerView-based chat interfaces.
- Database: Firebase Firestore for real-time cloud storage of messages, user profiles, and conversation metadata.
- Authentication: Firebase Authentication via email and password for user sign-in and registration.
- Push Notifications: Firebase Cloud Messaging (FCM) for real-time push notification delivery between users.
- AI Integration: Google Gemini API for natural language understanding and safety guidance within the Safe Hub feature.
- Deployment: Android APK, installable on Android devices without requiring specialised hardware or web hosting infrastructure.

Reading List

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